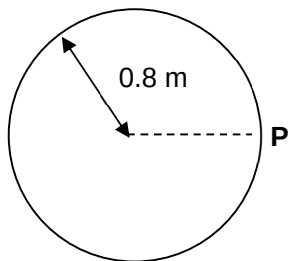


- 12 A model car moves in a circular path of radius 0.8 m at an angular speed of $0.5\pi \text{ rad s}^{-1}$.



What is its displacement from point P, 4 s after passing P?

- A** zero **B** 1.6 m **C** $0.8\pi \text{ m}$ **D** $1.6\pi \text{ m}$

Ans: A

$$v = r\omega = 0.8 \times \pi/2 = 0.4\pi,$$

$$s = vt = 0.4\pi \times 4 = 1.6\pi$$

$$\text{Circumference of circle} = 2\pi(0.8) = 1.6\pi$$

Therefore, displacement = 0

